HOLDER DIFFERENTIABILITY OF THE SEPARATING SINGULAR HYPERSURFACE FOR TRANSPORT INTO TWO CONVEX TARGETS WITHOUT SOURCE CONVEXITY

SHIBING CHEN, JIAKUN LIU

Editorial note. Source599 overstates affine invariance for arbitrary distance-to-boundary divided by diameter. The actual bound602 is determined by the collinear section endpoints explicitly specified519 and the ratio574-579, which is affine-invariant. The original formula is retained; the independent review accepts this uniquely determined local explanation without altering the PDE argument.

Editorial note. Source463 suppresses v(y_k) in its affine-function display;451 states the correct value and v(y_k) tends to v(p)=0, so the limiting argument is unchanged.

Editorial note. The old pdftex graphicx driver is omitted from the wrapper preamble; the existing XeLaTeX graphicx driver imports the unchanged original diagram. No author mathematical text is edited.

Editorial scope. One counted theorem t111, with all local supporting proofs353-678 as required solution. Standard prerequisites are only the named classical tools above; no local result is assumed. Requires independent substantive review before ready.

Allowed standard prerequisites (editorial).

- Brenier duality, interior Caffarelli regularity for convex targets, and the known Lipschitz separation of the two target regions. Use the transport potentials and initial Lipschitz hypersurface explicitly recalled in154-180 and303-316. No boundary Holder differentiability is given.. Br1,C92,CL1 Theorem5.1 and KM1 in retained author bibliography
- John normalization and standard centred-section properties. Centred convex sections exist, are dimensionally balanced, depend continuously on positive height, and satisfy the barrier estimate339.. C92b; CM AppendixA, particularlyA.8
- Localized Alexandrov estimates for Brenier solutions. Use la1/ua1 exactly as displayed365-374, with the upper estimate at a point whose subgradient meets an interior ball of the target. Constants may depend on the distance of that subgradient point to the target boundary. Also allow the

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same classical interior strict-convexity/differentiability result at a point inside the source whose subgradient meets the target interior, as invoked394 and632; this is Figalli-Kim Proposition3.4 in the author PDF (Proposition2 in the journal numbering).. Figalli-Kim2010, author version Proposition3.3 and equation3.1, PDF pages3-6; the paper cites final numbering Proposition1/equation4. Exact hypotheses read at https://people.math.ethz.ch/~afigalli/papers-pdf/Partial-regularity-of-Brenier-solutions-of-the-Monge-Ampere-equation.pdf

- Classical localized boundary uniform convexity and convex duality. Once a convex support has two-sided Alexandrov density and uniformly localized centred sections near a boundary point, the local Caffarelli boundary method yields a power lower modulus for its gradient; Legendre duality gives Holder continuity of the inverse gradient.. C96 Lemma2.2/Corollary2.3 and CM Theorem7.13, as invoked665-669. Applicability must be assessed; no local differentiability or localization lemma of the counted article is supplied freely.

Let $f, g \in L^1(\mathbb{R}^n)$ represent two probability densities concentrated on bounded open sets $\Omega, \Omega^* \subset \mathbb{R}^n$, respectively. Suppose there exists a positive constant λ such that $\frac{1}{\lambda} < f, g < \lambda$ in Ω, Ω^*, respectively. Invoking Brenier's theorem [1], we identify two globally Lipschitz convex functions, u and v, both defined on $\mathbb{R}^n$, satisfying the following conditions:

$$(Du)_\sharp f = g, \text{ with } Du(x) \in \overline{\Omega^*} \text{ for almost every } x \in \mathbb{R}^n; \tag{0.1}$$

$$(Dv)_\sharp g = f, \text{ with } Dv(y) \in \overline{\Omega} \text{ for almost every } y \in \mathbb{R}^n. \tag{0.2}$$

From the regularity theory of Caffarelli [2], if Ω^* is convex, then the potential function u belongs to $C^{1,\beta}_{\text{loc}}(\Omega)$ for some $\beta \in (0,1)$ and is strictly convex inside Ω.

If Ω^* consists of two disjoint convex domains, as seen in the example of [2] when mapping the unit disc onto two shifted half-discs, there appears a singular set $\mathcal{F} \subset \Omega$ on which u is not differentiable. In this paper, considering the case when $\Omega^* = \Omega_1^* \cup \Omega_2^*$, with Ω_1^* and Ω_2^* being bounded convex domains separated by a hyperplane $\mathcal{H}$, we will establish the regularity of the singular set $\mathcal{F}$.

From previous works [7, 17], one knows $\mathcal{F} \subset \Omega$ is a Lipschitz hypersurface, partitioning Ω into two subdomains Ω_i, $i = 1, 2$, such that

$$(Du)_\sharp(f\chi_{\Omega_i}) = g\chi_{\Omega_i^*} \qquad \text{for} \quad i = 1, 2. \tag{0.3}$$

Moreover, $\mathcal{F}$ is characterised as $\partial\Omega_i \cap \Omega$ for $i = 1, 2$, which can be expressed as the graph of a certain Lipschitz function.

Define the functions $u_i : \mathbb{R}^n \to \mathbb{R}$, for $i = 1, 2$, as follows:

$$u_i(x) := \sup\{L(x) : L \text{ is affine, } L \le u \text{ in } \Omega_i, \text{ and } DL(x) \in \Omega_i^*\} \quad \forall x \in \mathbb{R}^n. \tag{0.4}$$

From [7, 17], one has $\mathcal{F} = \{u_1 = u_2\} \cap \Omega$. Hence, if u_i are differentiable at $x \in \mathcal{F}$, the unit normal vector to the hypersurface $\mathcal{F}$ at x satisfies

$$\nu_{\mathcal{F}}(x) = \frac{Du_1(x) - Du_2(x)}{|Du_1(x) - Du_2(x)|}. \tag{0.5}$$

1. Preliminaries

Given a convex function $w : \mathbb{R}^n \to \mathbb{R}$, its subdifferential at $x \in \mathbb{R}^n$ is defined as:

$$\partial^- w(x) := \{y \in \mathbb{R}^n : w(z) \geq w(x) + y \cdot (z - x) \text{ for all } z \in \mathbb{R}^n\}. \tag{1.1}$$

For any subset $A \subset \mathbb{R}^n$, $\partial^- w(A) := \cup_{x \in A} \partial^- w(x)$.

Let $U \subset \mathbb{R}^n$ be an open set and C be a positive constant. We say that a convex function w satisfies the Monge-Ampère inequality:

$$\frac{1}{C}\chi_U \leq \det D^2 w < C\chi_U \tag{1.2}$$

in the Alexandrov sense if and only if for any Borel set A,

$$\frac{1}{C}|A \cap U| \leq |\partial^- w(A)| \leq C|A \cap U|, \tag{1.3}$$

where $|\cdot|$ denotes the n-dimensional volume.

Now, let us consider the functions and sets u, v, u_i and Ω_i, Ω_i^* for $i = 1, 2$ as delineated in the introduction. Specifically, we have the conditions (0.1)–(0.3) and u_i satisfies (0.4). It is important to note that Ω can be expressed as the union $\Omega = \Omega_1 \cup \Omega_2 \cup \mathcal{F}$, and within each Ω_i, $i = 1, 2$, we have $u_i = u$. Additionally, we have the following dual relationships:

$$v(y) = \sup_{x \in \Omega}\{y \cdot x - u(x)\} \quad \text{for all } y \in \mathbb{R}^n; \tag{1.4}$$

$$u_i(x) = \sup_{y \in \Omega_i^*}\{y \cdot x - v(y)\} \quad \text{for all } x \in \mathbb{R}^n, \ \ i = 1, 2. \tag{1.5}$$

Given that Ω_i^* are convex for $i = 1, 2$, Caffarelli's regularity theory [2] ensures that $u_i \in C^{1,\beta}_{\text{loc}}(\Omega_i)$ for some $\beta \in (0, 1)$, and they satisfy the Monge-Ampère inequality in the sense of Alexandrov:

$$C^{-1}\chi_{\Omega_i} \leq \det D^2 u_i \leq C\chi_{\Omega_i} \quad \text{for } i = 1, 2. \tag{1.6}$$

Lastly, we note that $\mathcal{F} = \{x \in \Omega : u_1(x) = u_2(x)\}$.

To proceed further, we introduce some useful definitions and notations. Let $w : \mathbb{R}^n \to \mathbb{R}$ be a convex function whose graph does not contain any infinite straight line.

Definition 1.1. *Given a point $y_0 \in \mathbb{R}^n$ and a small positive constant h, we define the centred sections of the convex function w at y_0 with height h as:*

$$S_h^c[w](y_0) := \{y \in \mathbb{R}^n : w(y) < w(y_0) + (y - y_0) \cdot \bar{p} + h\}, \tag{1.7}$$

where $\bar{p} \in \mathbb{R}^n$ is chosen such that the centre of mass of $S_h^c[w](y_0)$ coincides with y_0. Additionally, we define the sub-level set of w at y_0 with height h as:

$$S_h[w](y_0) := \{y \in \mathbb{R}^n : w(y) < \ell_{y_0}(y) + h\}, \tag{1.8}$$

where ℓ_{y_0} denotes an affine supporting function of w at y_0.

Remark 1.1. For the existence of $\bar{p}$ in (1.7), we refer the reader to [3]. If $w(0) = 0$ and $w \geq 0$, by [10, Remark 2.5], we have that

$$w \leq Ch \quad \text{in } S_h^c[w](0), \tag{1.9}$$

where C is a constant depending only on n.

In subsequent discussions, a constant will be referred to as universal if it depends solely on the parameters $n, \Omega, \Omega^*, \lambda$, and $\operatorname{dist}(x_0, \partial\Omega)$, where x_0 is a generic point of $\mathcal{F}$. Additionally, we will employ the notation C_n to represent a positive constant dependent exclusively on the dimension n, with the acknowledgment that its value may change across different contexts. The vectors e_i, where $i = 1, \ldots, n$, will be used to denote the standard basis vectors in the coordinate system.

Theorem 1.1. *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain, and let $\Omega^* = \Omega_1^* \cup \Omega_2^*$ be a union of bounded convex domains Ω_1^* and Ω_2^* in $\mathbb{R}^n$, separated by a hyperplane $\mathcal{H}$. If $\frac{1}{\lambda} < f, g < \lambda$ within Ω and Ω^*, respectively, for some positive constant λ, then the hypersurface $\mathcal{F}$ (as long as it is non-empty) is $C^{1,\beta}$ regular inside Ω for some $\beta \in (0,1)$. Additionally, $u_i \in C^{1,\beta}_{loc}(\Omega \cap \overline{\Omega_i})$ for $i = 1, 2$.*

2. $C^{1,\beta}$ regularity of $\mathcal{F}$.

In this section, we continue to utilise the notations established in Section 1 for $\Omega, \Omega^*, \Omega_i^*$ $(i = 1, 2)$, $u, v, u_i, \mathcal{F}$. Consider a specific point $x_0 \in \mathcal{F} \subset \Omega$. Select sequences $\{x_k\} \subset \Omega_1$ and $\{\hat{x}_k\} \subset \Omega_2$, both converging to x_0 as $k \to \infty$. By applying a translation of coordinates, we may assume $x_0 = 0$. Without loss of generality, and potentially passing to subsequences, we assume $\lim_{k\to\infty} Du(x_k) = y_0 \in \overline{\Omega_1^*}$, and $\lim_{k\to\infty} Du(\hat{x}_k) = \hat{y}_0 \in \overline{\Omega_2^*}$, respectively. This ensures that $y_0, \hat{y}_0 \in \partial^- u(0)$. By (1.4), we deduce that $0 \in \partial^- v(y_0) \cap \partial^- v(\hat{y}_0)$.

By subtracting an appropriate constant, we may assume $v(y_0) = v(\hat{y}_0) = 0$ and $v \geq 0$. Define $\Sigma = \{y \in \mathbb{R}^n : v(y) = 0\}$ and denote the set of extreme points of Σ as $\operatorname{ext}(\Sigma)$.

Recall Equation (0.2), especially $Dv(y) \in \overline{\Omega}$ for almost every $y \in \mathbb{R}^n$. Consider a section of v, namely $Z := \{y \in \mathbb{R}^n : v(y) < \ell(y)\}$, with ℓ being an affine function. Provided that Z is bounded and contains a point y such that $0 \in \partial^- v(y)$, since $0 \in \mathcal{F}$ is inside the interior of Ω, we can assert that $\partial^- v(y) \cap \Omega \neq \emptyset$. Denoting the centre of mass of Z by z, and employing John's lemma, we identify an affine transformation L such that $L(z) = 0$ and $B_1(0) \subset LZ \subset B_{C_n}(0)$, where C_n solely depends on the dimension n.

From [14, Inequality (4)], we deduce that:

$$|\inf_Z (v - \ell)|^n \geq C_1 \left| \left(\frac{1}{2} Z\right)_z \cap \Omega^* \right| |Z|, \tag{2.1}$$

where $\left(\frac{1}{2}Z\right)_z$ represents the dilation of Z with respect to z, and C_1 is a constant depending on n and λ. From [14, Proposition 1], we have the following bound:

$$|v(y) - \ell(y)|^n \leq C_2 |Z \cap \Omega^*| |Z| \operatorname{dist}\big(L(y), \partial(L(Z))\big) \quad \text{for } y \in Z, \tag{2.2}$$

where C_2 is a constant depending only on n, λ, and the diameter of Ω.

Lemma 2.1. *The set Σ is both bounded and closed, and $\operatorname{ext}(\Sigma) \subset \partial\Omega^*$.*

Proof. First, we prove that Σ is bounded and closed. Assume, to the contrary, that this is not the case. By convexity, Σ would then contain a half-line of the form $\{q + te : t \geq 0\}$ for some $q \in \Sigma$ and $e \in \mathbb{S}^{n-1}$. For any $y \in \mathbb{R}^n$ and $x \in \partial^- v(y)$, by the convexity of v one has

$$(x - 0) \cdot (y - q - te) \geq 0.$$

As t tends to infinity, this implies that $x \cdot e \leq 0$. Given the arbitrariness of y, it follows that

$$\partial^- v(\mathbb{R}^n) \subseteq \{x \cdot e \leq 0\}.$$

Since we have $(Dv)_\sharp g = f$, it can be deduced that $\Omega \subseteq \partial^- v(\mathbb{R}^n) \subseteq \{x \cdot e \leq 0\}$, which contradicts the fact that $0 \in \Omega$. Thus, Σ must be bounded and closed.

Next, we prove that $\text{ext}(\Sigma) \subseteq \partial\Omega^*$. Suppose to the contrary that there exists a point $p \in \text{ext}(\Sigma)$ such that p is either in Ω^* or in $\mathbb{R}^n \setminus \overline{\Omega^*}$. If $p \in \text{ext}(\Sigma) \cap \Omega^*$, then we have $0 \in \partial^- v(p) \cap \Omega$. Referring to Proposition 2 and the subsequent discussion in [14], v is inferred to be C^1 and strictly convex in a neighborhood of p. Consequently, by duality, the function u must be differentiable at 0, leading to a contradiction. In the case where $p \in \text{ext}(\Sigma) \cap (\mathbb{R}^n \setminus \overline{\Omega^*})$, the application of inequalities (2.1) and (2.2), along with the arguments presented in Section 3.1 of [14], also yields a contradiction. Hence, we conclude that $\text{ext}(\Sigma) \subset \partial\Omega^*$. □

Since exposed points are dense in the set of extreme points, we can choose points

$$p \in \text{ext}(\Sigma) \cap \partial\Omega_1^* \quad \text{and} \quad \hat{p} \in \text{ext}(\Sigma) \cap \partial\Omega_2^* \tag{2.3}$$

such that both are exposed points of Σ. In this context, we establish the following localisation properties.

Lemma 2.2. *Given any $r_0 > 0$, there exists an $h_0 > 0$ such that for all $h \leq h_0$, the inclusion $S_h^c[v](p) \subset B_{r_0}(p)$ holds, where p is as in* (2.3).

Proof. Since p is an exposed point of Σ, there exists a unit vector $e \in \mathbb{S}^{n-1}$ such that

$$\Sigma \subset \{y : (y - p) \cdot e \leq 0\},$$
$$\Sigma \cap \{y : (y - p) \cdot e = 0\} = \{p\}.$$

Since $S_h^c[v](p)$ is centred at p, if $S_h^c[v](p) \cap \{y : (y - p) \cdot e \geq 0\} \subset B_r(p)$, then it follows that $S_h^c[v](p) \subset B_{C_n r}(p)$.

Observing that $v > 0$ in $\mathbb{R}^n \setminus \Sigma$ and given any small constant $r_0 > 0$, the convexity and continuity of v ensure the existence of a constant δ_0 such that

$$v > \delta_0 \quad \text{in} \quad \{y : (y - p) \cdot e \geq 0\} \setminus B_{\frac{r_0}{C_n}}(p). \tag{2.4}$$

By (1.9) we have $v \leq C_n h$ in $S_h^c[v](p)$. Setting $h_0 = \frac{\delta_0}{C_n}$, for all $h \leq h_0$, we then have $v \leq \delta_0$ in $S_h^c[v](p)$. From (2.4), it follows that

$$S_h^c[v](p) \cap \{y : (y - p) \cdot e \geq 0\} \subset B_{\frac{r_0}{C_n}}(p) \quad \forall\ h \leq h_0.$$

Therefore, we conclude that $S_h^c[v](p) \subset B_{r_0}(p)$ whenever $h \leq h_0$. □

The next lemma establishes the localisation of centred sections near the point p.

Lemma 2.3. *For any given $\frac{1}{2} > r_0 > 0$, let h_0 be the constant specified in Lemma 2.2. There exists a positive constant $\delta_0 > 0$ such that for all points $y \in \overline{\Omega_1^*} \cap B_{\delta_0}(p)$ and for all $h \le h_0$, the inclusion $S_h^c[v](y) \subset B_{2r_0}(p)$ holds, where p is as in* (2.3).

Proof. Assume, for the sake of contradiction, that the statement is false. Then, there exist sequences $\{y_k\} \subset \overline{\Omega_1^*}$ converging to p and $\{h_k\}$ with $h_k \le h_0$, such that

$$S_{h_k}^c[v](y_k) \cap (\mathbb{R}^n \setminus B_{2r_0}(p)) \neq \emptyset. \tag{2.5}$$

Given that $S_{h_k}^c[v](y_k)$ is balanced with respect to y_k, there exists a line segment $I_k \subset S_{h_k}^c[v](y_k)$ centred at y_k satisfying $|I_k| \ge C_n r_0$. Without loss of generality, by taking a subsequence if necessary, we can assume that $h_k \to \bar{h} \le h_0$ and $I_k \to I_\infty$ as $k \to \infty$, where I_∞ is a segment centred at p and satisfying

$$|I_\infty| \ge C_n r_0. \tag{2.6}$$

First, we claim that $S_{h_k}^c[v](y_k)$ is uniformly bounded. From Lemma 2.1, Σ is compact. Owing to the non-negativity and convexity of v, we deduce the existence of a large constant $R > 1$ satisfying

$$v(y) > h_0 + \sup_{B_1(p)} v \quad \forall\, y \in \mathbb{R}^n \setminus B_R(p).$$

Suppose $S_{h_k}^c[v](y_k) = \{y \in \mathbb{R}^n : v(y) < \ell_k\}$, for some affine function ℓ_k that fulfils $\ell_k(y_k) = v(y_k) + h_k$. For any given unit vector $e \in \mathbb{S}^n$, we may assume ℓ_k is decreasing in the e-direction, otherwise consider $-e$ instead.

Since y_k converges to p, we may assume that $y_k \in B_1(p)$ for all k. It can be verified that $y_k + 3Re \in \mathbb{R}^n \setminus B_R(p)$, leading to the inequality

$$v(y_k + 3Re) > h_0 + \sup_{B_1(p)} v \ge \ell_k(y_k) \ge \ell_k(y_k + 3Re).$$

This implies $y_k + 3Re \notin S_{h_k}^c[v](y_k)$. Since $S_{h_k}^c[v](y_k)$ is centred at y_k, we deduce $y_k - 3C_nRe \notin S_{h_k}^c[v](y_k)$. Given that e is arbitrary, we conclude the claim that

$$S_{h_k}^c[v](y_k) \subset B_{3C_nR}(y_k) \subset B_{3C_nR+1}(p). \tag{2.7}$$

Now, let's consider a subsequence (without changing notation) such that $S_{h_k}^c[v](y_k)$ converges to a limit convex set S_∞ in the Hausdorff distance, and $h_k \to \bar{h} \le h_0$ as $k \to \infty$. Write $\ell_k = x_k \cdot (y - y_k) + h_k$ with $x_k \in \mathbb{R}^n$, the global Lipschitz continuity of v ensures that $\sup_k |x_k| \le \|v\|_{\mathrm{Lip}} < \infty$. Consequently, after possibly taking another subsequence, we can assert that $x_k \to \bar{x}$ as $k \to \infty$.

In the case when $\bar{h} > 0$, it follows that

$$S_\infty = \left\{y \in \mathbb{R}^n : v(y) < \bar{x} \cdot (y - p) + \bar{h}\right\} = S_{\bar{h}}^c[v](p).$$

By (2.5), passing to limit we have

$$S_{\bar{h}}^c[v](p) \cap (\mathbb{R}^n \setminus B_{2r_0}(p)) \neq \emptyset,$$

leading to a contradiction with Lemma 2.2.

In the case when $\bar{h} = 0$, we observe that $0 \leq v \leq \bar{x} \cdot (y - p)$ for all y in I_∞. Given that I_∞ is centred at p, it follows that v must be identically zero on I_∞, implying that $I_\infty \subseteq \Sigma$. This contradicts to the fact that p is an exposed point of Σ. □

Take r_0 small so that $B_{2r_0}(p) \cap \Omega_2^* = \emptyset$. Let h_0, δ_0 be as in Lemma 2.2 and 2.3. Define $\rho_0 := \frac{1}{2}\text{dist}(0, \partial\Omega)$. We will show the differentiability of v near p in the following lemma.

Lemma 2.4. *There exists a constant $\bar{r}_1 < \frac{1}{3}\delta_0$, such that v is differentiable at all $y \in \overline{\Omega_1^*} \cap B_{\bar{r}_1}(p)$, where p is as in* (2.3).

Proof. First, we establish the differentiability of v at the point p. Before giving the detailed proof, let us outline the main idea. Suppose v is not differentiable at p. By subtracting an affine function, we may assume that v satisfies (2.8) for some unit vector e pointing outside Ω_1^*. Consider the section $S_h^c[v](y_{t_0}) = \{v < \ell\}$, centred at $y_{t_0} := p - t_0 e \in \Omega_1^*$, where $t_0 > 0$ is chosen small so that y_{t_0} lies sufficiently close to p. We have two observations: (i) p is nearly the minimum point of $v - \ell$, and (ii) After applying a normalisation via the affine transformation L in (2.19), $L(p)$ lies very close to the boundary of $L\left(S_h^c[v](y_{t_0})\right)$ as $h \to 0$. Using the estimates in (2.1) and (2.2), we arrive at a contradiction. This argument resembles Caffarelli's localisation technique and the discussion around equations (1.4) and (1.5) in [15], but involves new delicate estimates as $p \in \partial\Omega_1^*$ lies on the boundary. Below, we provide the detailed proof.

Consider n linearly independent unit vectors $\hat{e}_i$ (may not be orthogonal), where $i = 1, \ldots, n$. By convexity of v, it suffices to show that $v(p + t\hat{e}_i)$ is differentiable at $t = 0$ as a convex function of a single variable t, for each $i = 1, \ldots, n$.

Suppose to the contrary that v is not differentiable at p. Then there exists a unit vector e, such that $\{p + te : t < 0\} \cap \Omega_1^* \neq \emptyset$, and $v(p + te)$ is not differentiable at $t = 0$. By subtracting an affine function ℓ_1, we can assume the following behaviour (see Figure 3.1):

$$\begin{aligned} v &\geq 0 && \text{on } \mathbb{R}^n, \\ v(p + te) &= o(t) && \text{for } t < 0, \\ v(p + te) &= at + o(t) && \text{for } t > 0, \end{aligned} \tag{2.8}$$

where $a > 0$ is a constant. Note that the target of Dv is changed to $\tilde{\Omega} = \Omega - \{D\ell_1\}$, a translation of Ω. Given that $\partial^- v(p) \cap \tilde{\Omega}$ is non-empty, the estimates (2.1)–(2.2) still apply.

Define $y_{t_0} := p - t_0 e$ for $t_0 \in (0, \frac{1}{2}\delta_0)$ sufficiently small. Let's consider the centred section

$$S_h^c[v](y_{t_0}) = \{v < \ell\},$$

where ℓ is an affine function satisfying $\ell(y_{t_0}) = v(y_{t_0}) + h$. Consider the intersection of the boundary $\partial S_h^c[v](y_{t_0})$ with the line $\{p + te : t \in \mathbb{R}\}$, and let

$$\partial S_h^c[v](y_{t_0}) \cap \big\{p + te : t \in \mathbb{R}\big\} = \big\{p - t_1(h)e, \quad p + t_2(h)e\big\},$$

where $t_1(h) > t_0$ and $t_2(h) > -t_0$. By [5, Lemma A.8], we have that $S_h^c[v](y_{t_0})$ varies continuously with respect to $h \in (0, h_0]$. Hence, $t_1(h)$ and $t_2(h)$ depend

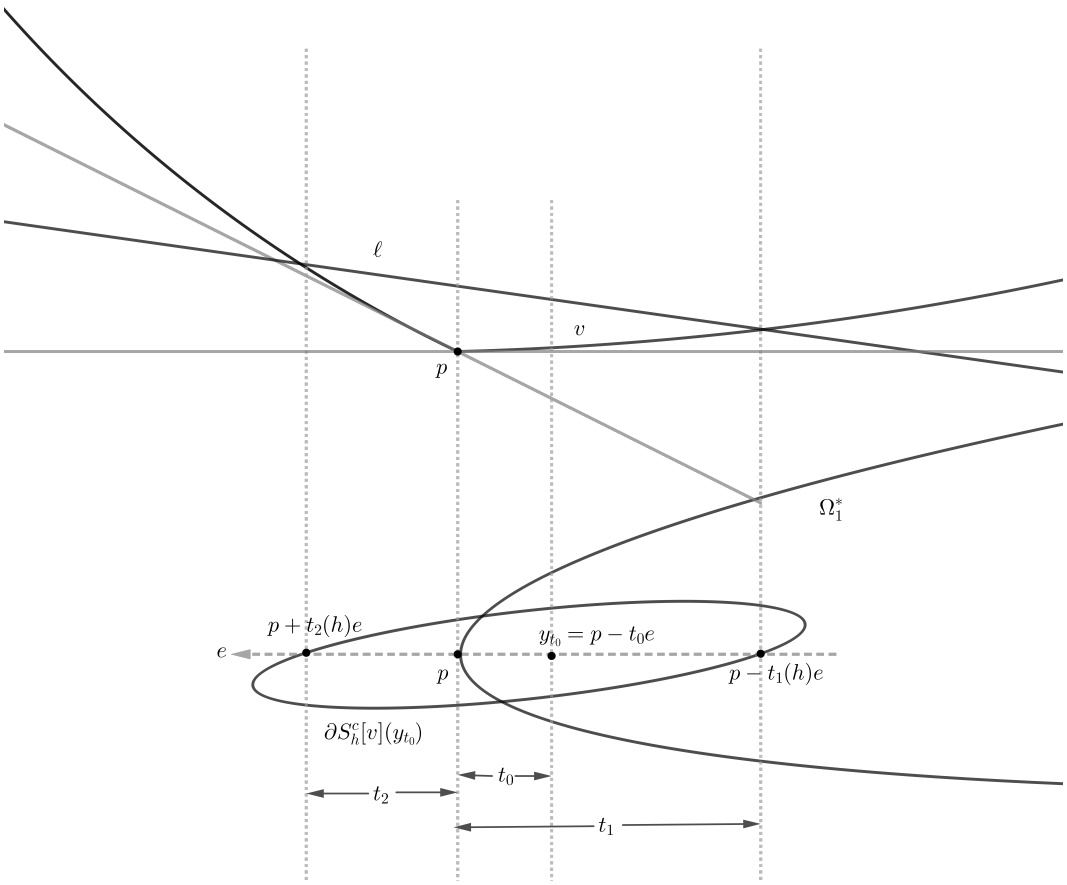


FIGURE 2.1. A centred section $S_h^c[v](y_{t_0})$, near p.

continuously on h. Since $S_h^c[v](y_{t_0})$ is centred at y_{t_0}, we have the following bounds:

$$t_0 < t_1(h) < C_n(t_0 + t_2(h)), \tag{2.9}$$

where C_n is a constant depending only on n.

We first *claim* that $t_2(h_0) \geq t_0$, provided t_0 is chosen sufficiently small initially. Suppose, to the contrary, that $t_2(h_0) < t_0$. Then, by (2.8), we have

$$\ell(p + t_2(h_0)e) = v(p + t_2(h_0)e) \leq 2at_0, \tag{2.10}$$

provided t_0 is small enough. By (2.8) and (2.9), we obtain

$$\ell(p - t_1(h_0)e) = v(p - t_1(h_0)e) = o(t_0). \tag{2.11}$$

By (2.10), (2.11), and the balance of $S_{h_0}^c[v](y_{t_0})$ with respect to y_{t_0}, we have

$$\ell(y_{t_0}) \leq C_n(2at_0 + o(t_0)),$$

contradicting the fact that $\ell(y_{t_0}) \geq \ell(y_{t_0}) - v(y_{t_0}) = h_0$, provided $t_0 \ll h_0$ was chosen initially. Hence, the claim is proved.

We then *claim* that $\liminf_{h\to 0} t_2(h) \leq 0$, provided t_0 is chosen sufficiently small initially. Suppose, to the contrary, that there exists a constant $a_0 \in (0, t_0)$ and a

sequence $\{h_j\}$ converging to 0, such that $t_2(h_j) \geq a_0$ for all j. Hence, from (2.8),

$$v(p + t_2(h_j)e) \geq \frac{1}{2}aa_0, \tag{2.12}$$

provided t_0 is small enough. Moreover, since $S^c_{h_j}[v](y_{t_0})$ is centred at y_{t_0} we have the segment

$$I = \left\{p + \theta e : -t_0 - \frac{1}{C_n}t_0 < \theta < a_0\right\} \subset S^c_{h_j}[v](y_{t_0}) \quad \forall\, j = 1, 2, \cdots . \tag{2.13}$$

Suppose $S^c_{h_j}[v](y_{t_0}) = \{y \in \mathbb{R}^n : v(y) < v(y_{t_0}) + x_j \cdot (y - y_{t_0}) + h_j\}$ for some vector $x_j \in \mathbb{R}^n$. Since v is globally Lipschitz, $\sup_j |x_j| \leq \|v\|_{\text{Lip}} < \infty$. Consequently, by taking a subsequence if necessary, we may assume $x_j \to \bar{x}$ as $j \to \infty$. In the limit, we thus obtain

$$0 \leq v(y) \leq \bar{\ell}(y) \quad \forall\, y \in I,$$

where $\bar{\ell}(y) = v(y_{t_0}) + \bar{x} \cdot (y - y_{t_0})$. Since $y_{t_0} \in I$, $v(y_{t_0}) = \bar{\ell}(y_{t_0})$ and $v \leq \bar{\ell}$ on I, the convexity of v implies that $v \equiv \bar{\ell}$ on I, namely $v(p+te)$ is affine with respect to t for $t \in (-t_0 - \frac{1}{C_n}t_0, a_0)$, which contradicts (2.8). Hence, the claim $\liminf_{h\to 0} t_2(h) \leq 0$ is proved.

Now, choose a $t_0 > 0$ small enough such that the above two claims hold. By these claims and the continuity of $t_2(h)$ with respect to h, there exists an $h' \in (0, h_0)$ such that

$$t_2(h') = \frac{1}{M}t_0, \tag{2.14}$$

where $M > 1$ is a large constant to be determined. By (2.9), we then have

$$t_0 < t_1(h') < C_n(t_0 + t_2(h')) \leq 2C_n t_0. \tag{2.15}$$

Assume that $S^c_{h'}[v](y_{t_0}) = \{v < \ell\}$ for some affine function ℓ satisfying $\ell(y_{t_0}) = v(y_{t_0}) + h'$.

Evaluating the affine function ℓ at specific points yields:

$$\ell(p - t_1 e) = v(p - t_1(h')e) = o(t_1) = o(t_0), \tag{2.16}$$

$$\ell\left(p + t_2(h')e\right) = v\left(p + \frac{1}{M}t_0 e\right) = \frac{a}{M}t_0 + o(t_0). \tag{2.17}$$

The above equations imply that ℓ is increasing along the e direction (provided t_0 is chosen sufficiently small at the beginning), leading to:

$$\ell(p) - v(p) = \ell(p) \geq \ell(y_{t_0}) \geq \ell(y_{t_0}) - v(y_{t_0}) \approx \left|\inf_{S^c_{h'}[v](y_{t_0})}(v - \ell)\right|. \tag{2.18}$$

Applying John's lemma, we identify an affine transformation L such that $L(y_{t_0}) = 0$, and

$$B_1(0) \subset L\left(S^c_{h'}[v](y_{t_0})\right) \subset B_{C_n}(0). \tag{2.19}$$

Since the affine transformation preserves the ratio of distances, we have

$$\frac{\text{dist}\left(L(p), \partial L\left(S^c_{h'}[v](y_{t_0})\right)\right)}{\text{diam}(L\left(S^c_{h'}[v](y_{t_0})\right))} = \frac{\text{dist}(p, \partial S^c_{h'}[v](y_{t_0})}{\text{diam}(S^c_{h'}[v](y_{t_0})} \leq \frac{t_2(h')}{t_1(h')} \leq \frac{1}{M}.$$

Hence, by (2.19)

$$\text{dist}\left(L(p), \partial L\left(S^c_{h'}[v](y_{t_0})\right)\right) \leq \frac{C_n}{M}. \tag{2.20}$$

Since $y_{t_0} \in \Omega_1^* \cap B_{\delta_0}(p)$ and $h' \leq h_0$, Lemma 2.3 implies $S^c_{h'}[v](y_{t_0}) \subset B_{2r_0}(p)$, ensuring that $S^c_{h'}[v](y_{t_0}) \cap \Omega_2^* = \emptyset$. The convexity of Ω_1^* leads to the volume comparison:

$$\left| \left(\frac{1}{2} S^c_{h'}[v](y_{t_0}) \right)_{y_{t_0}} \cap \Omega^* \right| \geq C_n \left| S^c_{h'}[v](y_{t_0}) \cap \Omega^* \right|. \tag{2.21}$$

Then by the estimate (2.1), we can obtain

$$\left| \inf_{S^c_{h'}[v](y_{t_0})} (v - \ell) \right|^n \geq C_1 C_n |S^c_{h'}[v](y_{t_0}) \cap \Omega^*| |S^c_{h'}[v](y_{t_0})|, \tag{2.22}$$

where C_1 and C_n are constants independent of M.

On the other hand, by estimates (2.2), (2.18) and (2.20), we can deduce

$$\left| \inf_{S^c_{h'}[v](y_{t_0})} (v - \ell) \right|^n \lesssim |\ell(p) - v(p)|^n \leq \frac{C_2 C_n}{M} |S^c_{h'}[v](y_{t_0}) \cap \Omega^*| |S^c_{h'}[v](y_{t_0})|, \tag{2.23}$$

where C_2 is another constant independent of M. However, this contradicts the estimate (2.22), when the constant M is sufficiently large.

Consequently, we conclude that the function v is differentiable at the point p. By selecting a sufficiently small $\bar{r}_1 < \delta_0$, we further obtain:

$$\partial^- v(B_{\bar{r}_1}(p)) \subset B_{\rho_0}(0) \subset \Omega. \tag{2.24}$$

Last, for any point $y \in \overline{\Omega_1^*} \cap B_{\bar{r}_1}(p)$, we can employ the previous argumentation for establishing the differentiability of v at p to similarly show that v is differentiable at y. □

Remark 2.1. Let r_0, δ_0, ρ_0, and $\bar{r}_1$ be the constants given in preceding lemmas. According to (2.24), for any point $y \in \Omega_1^* \cap B_{\bar{r}_1}(p)$, the set $\partial^- v(y) \cap \Omega$ is non-empty. By invoking Proposition 2 from [14] and the subsequent discussions, it follows that the function v is continuously differentiable and exhibits strict convexity within $\Omega_1^* \cap B_{\bar{r}_1}(p)$.

Lemma 2.5 (Localisation)**.** *The function v satisfies the Monge-Ampère inequality:*

$$\frac{1}{\lambda^2} \chi_{\Omega_1^*} \leq det D^2 v \leq \lambda^2 \chi_{\Omega_1^*} \quad \text{in } B_{\bar{r}_1}(p) \tag{2.25}$$

in the Alexandrov sense, as in (1.3)*.*

Proof. It suffices to show $\det D^2 v$ has no mass in $B_{\bar{r}_1}(p) \setminus \Omega_1^*$, namely $|\partial^- v(B_{\bar{r}_1}(p) \setminus \Omega_1^*)| = 0$.

In fact, let $y \in B_{\bar{r}_1}(p) \setminus \Omega_1^*$, and $x \in \partial^- v(y)$. Owing to (2.24), it is guaranteed that x belongs to $\Omega = \Omega_1 \cup \Omega_2 \cup \mathcal{F}$. Recall that u_i is C^1 and strictly convex inside Ω_i for $i = 1, 2$. If x were to lie in either of Ω_i, it would imply that $y \in \Omega_i^*$, being the gradient $Du_i(x)$. This, however, contradicts our initial assumption that $y \in B_{\bar{r}_1}(p) \setminus \Omega_1^*$. Consequently, it must be that x resides in the set $\mathcal{F}$.

This reasoning concludes that the subdifferential $\partial^- v$ maps $B_{\bar{r}_1}(p) \setminus \Omega_1^*$ entirely into $\mathcal{F}$. By [7, Theorem 5.1], we have that $\mathcal{F}$ is a Lipschitz hypersurface, and hence

its Lebesgue measure is zero. Therefore, the Lebesgue measure of the image set satisfies $|\partial^- v(B_{\bar{r}_1}(p) \setminus \Omega_1^*)| = 0$. □

Utilising Lemmas 2.2 and 2.3, we are able to determine constants $\bar{r}_0 < \bar{r}_1$ and $h_0 > 0$ such that:

$$S_h^c[v](y) \subset B_{\bar{r}_1}(p) \quad \forall\, y \in B_{\bar{r}_0}(p) \cap \Omega_1^*, \quad \forall\, h \le h_0. \tag{2.26}$$

With these preparations, we are now ready to prove Theorem 1.1 as follows.

Proof of Theorem 1.1. By (2.25) and (2.26), we can invoke the proof of [4, Lemma 2.2, Corollary 2.3] (see also [5, Theorem 7.13]) to obtain a quantitative uniform convexity of v within $B_{\bar{r}_0}(p) \cap \overline{\Omega_1^*}$. Specifically, this implies that for any two points $y, \tilde{y} \in B_{\bar{r}_0}(p) \cap \overline{\Omega_1^*}$, one has the inequality

$$|Dv(y) - Dv(\tilde{y})| \ge C|y - \tilde{y}|^a,$$

where $a > 2$ and C is a positive constant.

Since $u_1 = v^*$ in Ω_1, where v^* is the Legendre transform of v, a basic and well known property of the Legendre transform [5, Remark 7.10] allows us to deduce that $u_1 \in C^{1,\beta}(B_r(0) \cap \overline{\Omega_1})$ for some $\beta \in (0,1)$ and $r > 0$. By a similar argument, and a possibly smaller r, we can ensure that $u_2 \in C^{1,\beta}(B_r(0) \cap \overline{\Omega_2})$.

Inferred from (0.5), we conclude that the singular set $\mathcal{F}$ exhibits $C^{1,\beta}$ regularity in $B_r(0)$. Finally, a standard covering argument ensures the $C^{1,\beta}$ regularity of $\mathcal{F}$ inside Ω. □

Remark 2.2. Since $u_i \in C^{1,\beta}(B_r(0) \cap \overline{\Omega_i})$ for $i = 1, 2$, we have $p = y_0 = Du_1(0)$ and $\hat{p} = \hat{y}_0 = Du_2(0)$.

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